

gdTAI V4.2 NK and T lineage cluster review

This read-only review uses the completed development-only scVI representation. UMAP coordinates and marker plots use the deterministic 250,000-cell diagnostic sample; cluster cell and anchor counts use all 4,023,462 cells. No locked cohort was included.

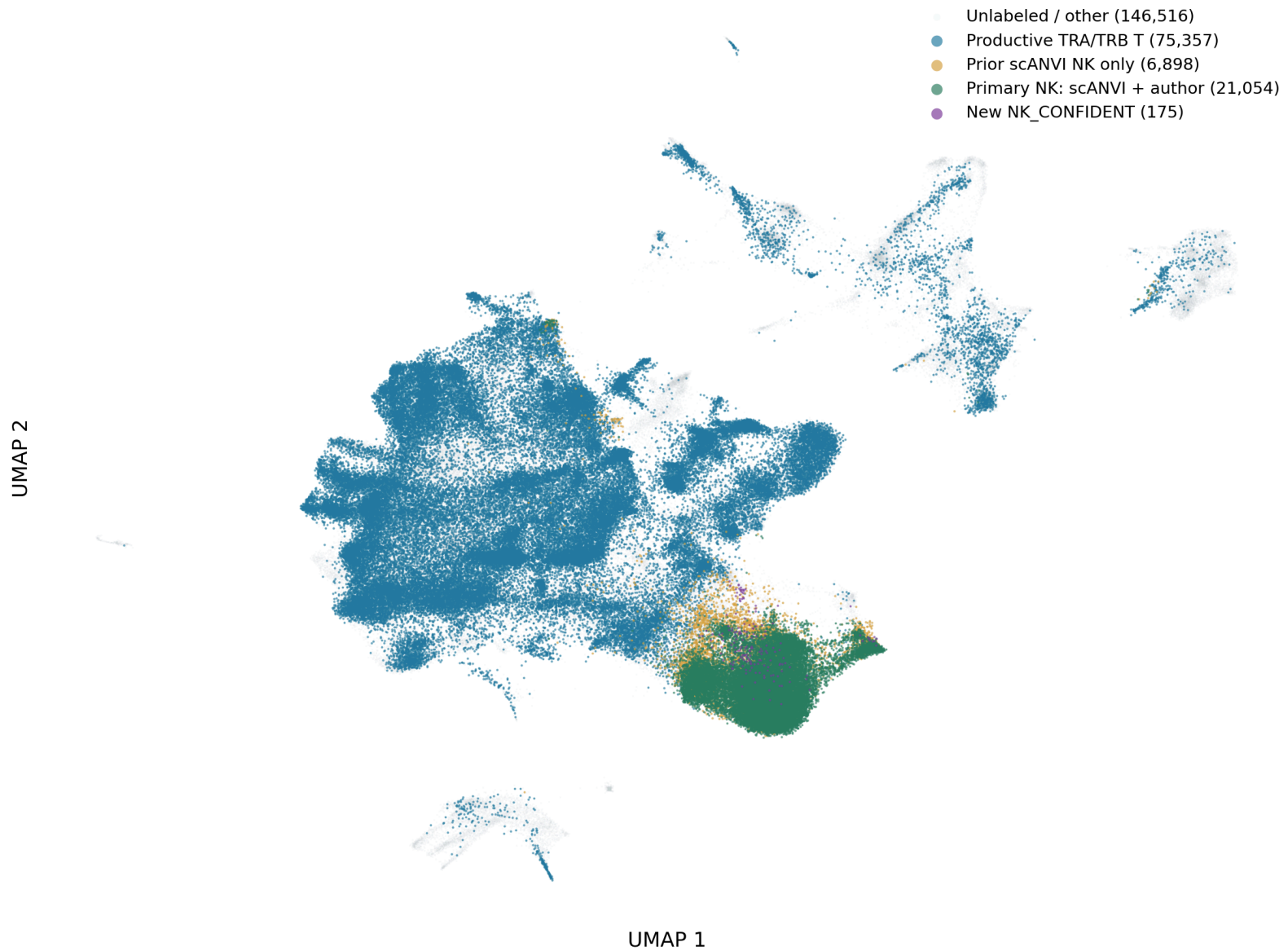
Result: PASS_VISUAL_REVIEW_READY. The feature plots, recoverable annotation evidence, and independent Leiden clusters are ready for biological review before changing the NK label definition.

24 unsupervised clusters	21,054 primary NK anchors	967,149 restored productive-T anchors	469 current NK_CONFIDENT cells
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1. Annotation evidence on the scVI UMAP

The missing historical integrated H5AD prevents recovery of its complete full-cell scANVI annotation column. The plot therefore shows only traceable evidence: prior scANVI-NK cells retained in the frozen V4 manifest, dual scANVI-plus-author NK anchors, restored productive-TRA/TRB T anchors, and the 469 current two-evidence NK candidates.

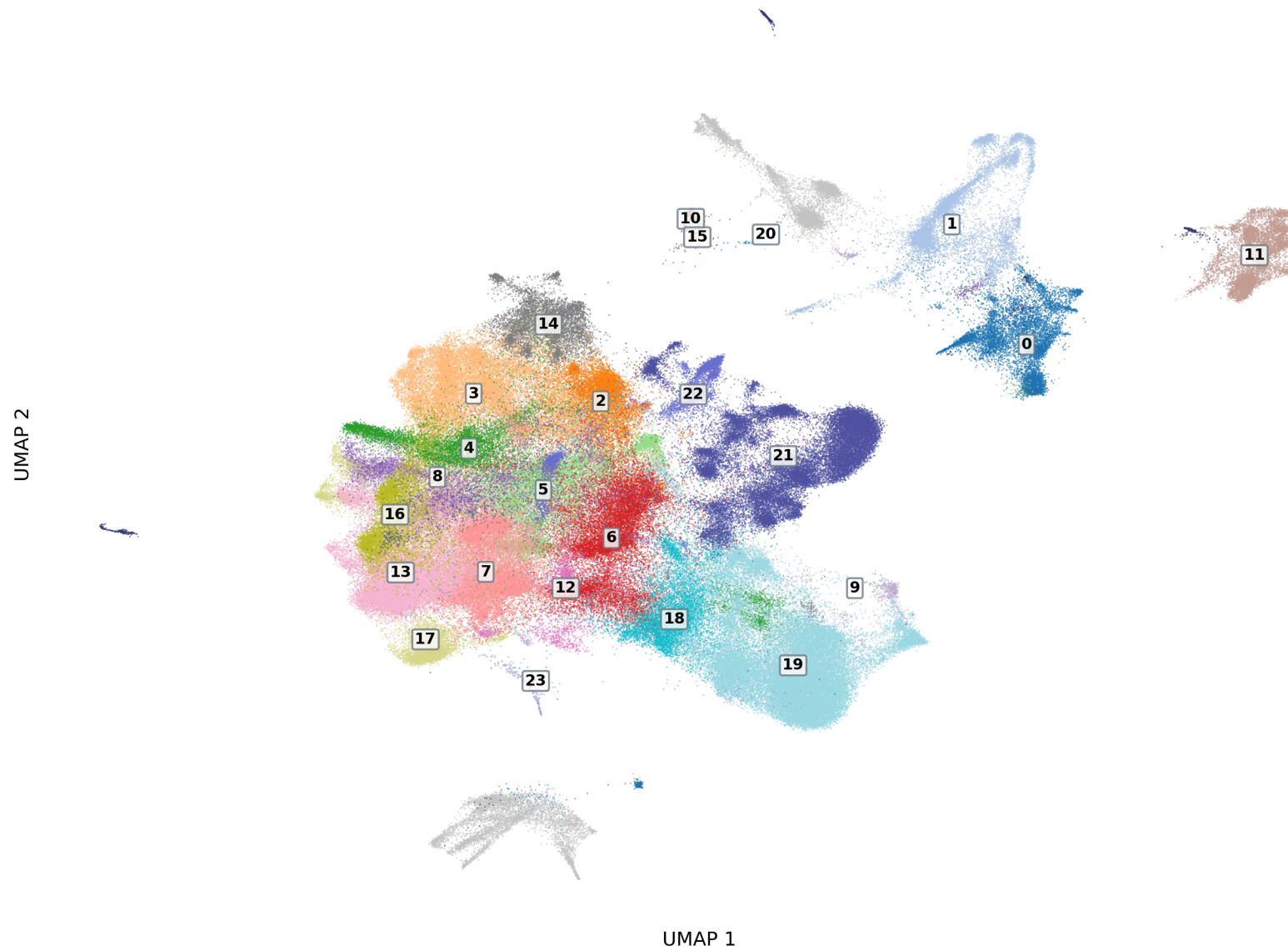
Recoverable prior annotation evidence on the scVI UMAP



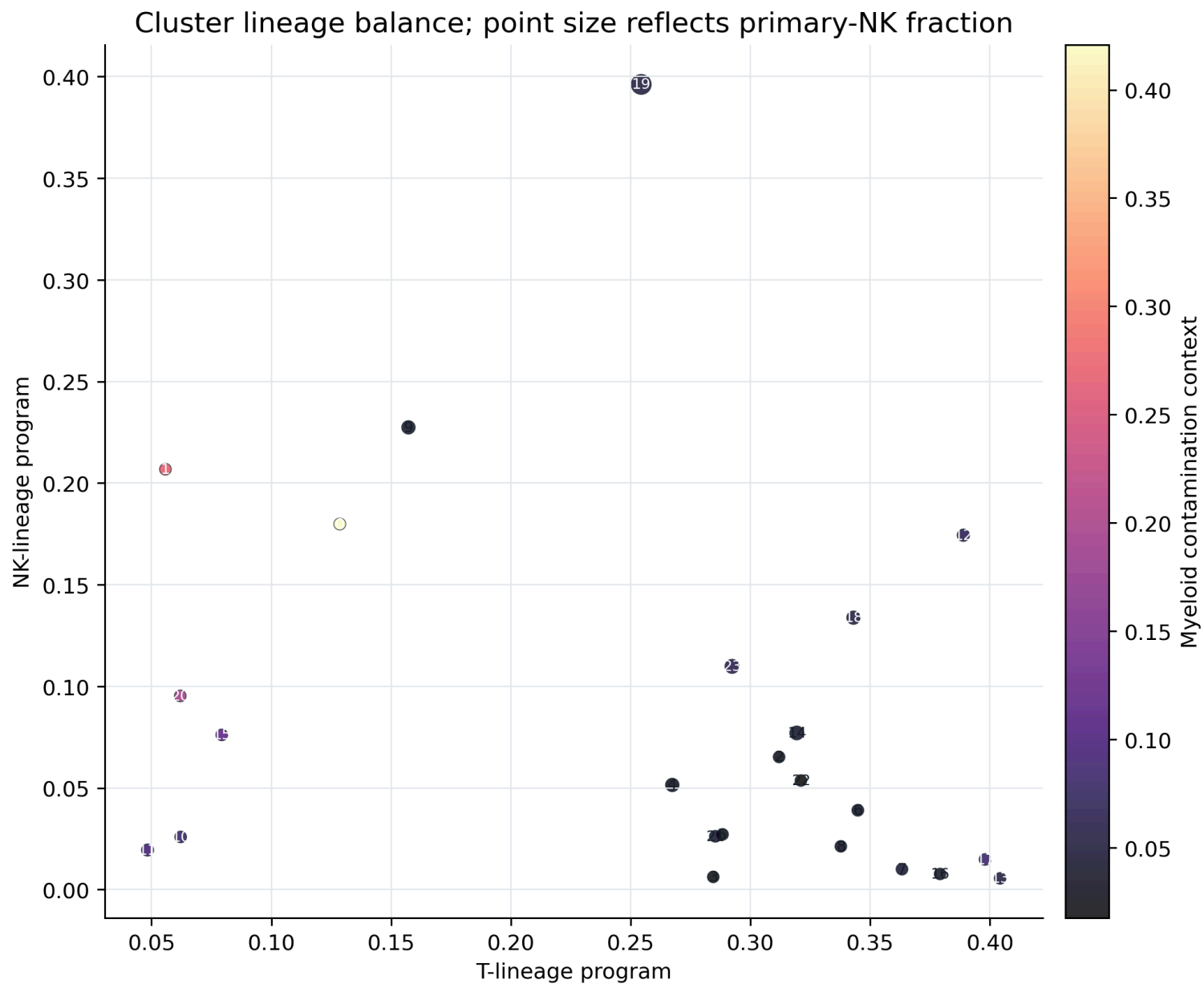
These are provenance classes, not a new annotation model.

2. Unsupervised structure

Unsupervised scVI Leiden clusters: global_r0.8_s29



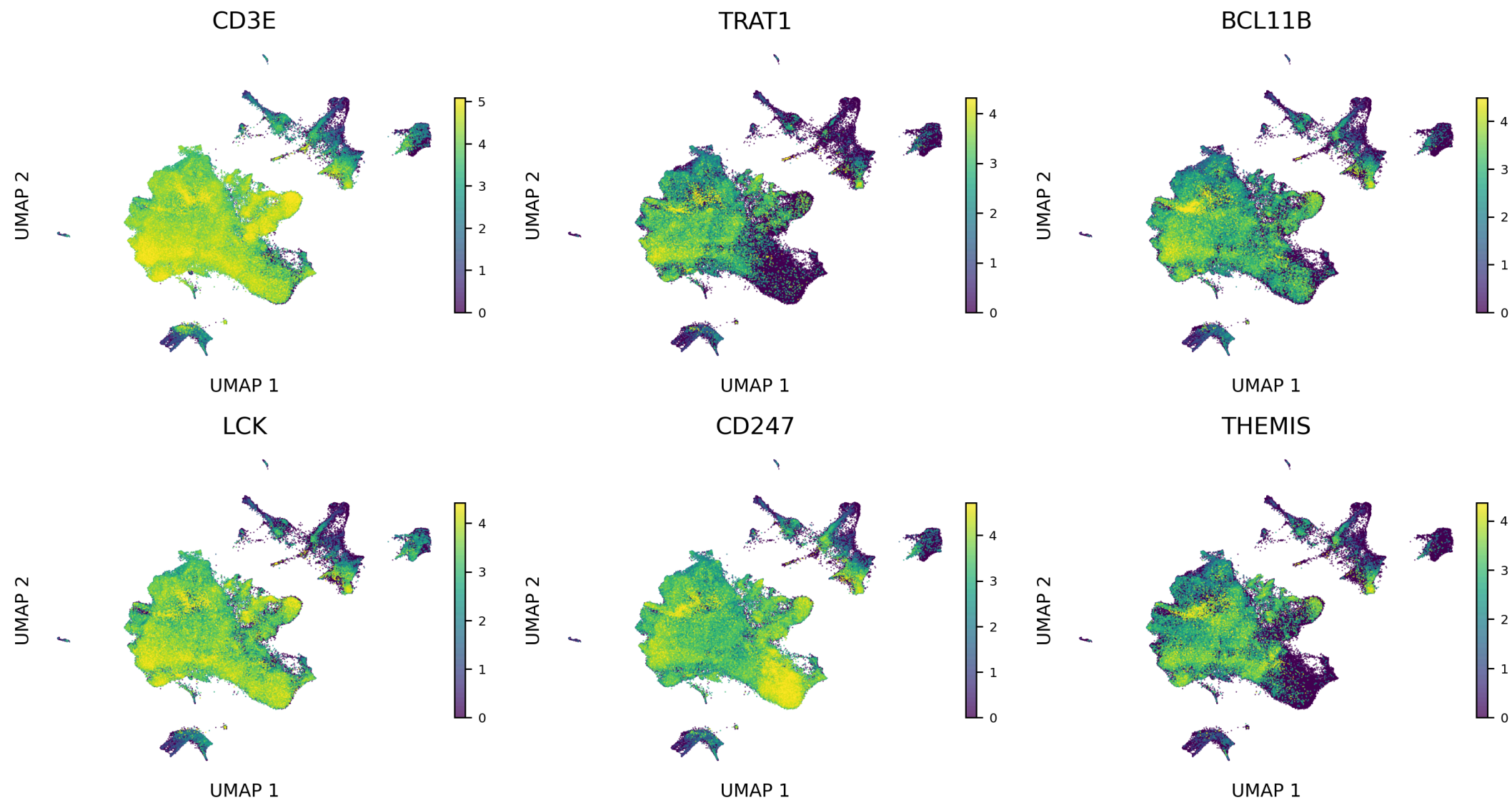
Frozen global Leiden partition `global_r0.8_s29`.



Scores combine multiple genes; color shows myeloid contamination context and point size shows primary-NK fraction.

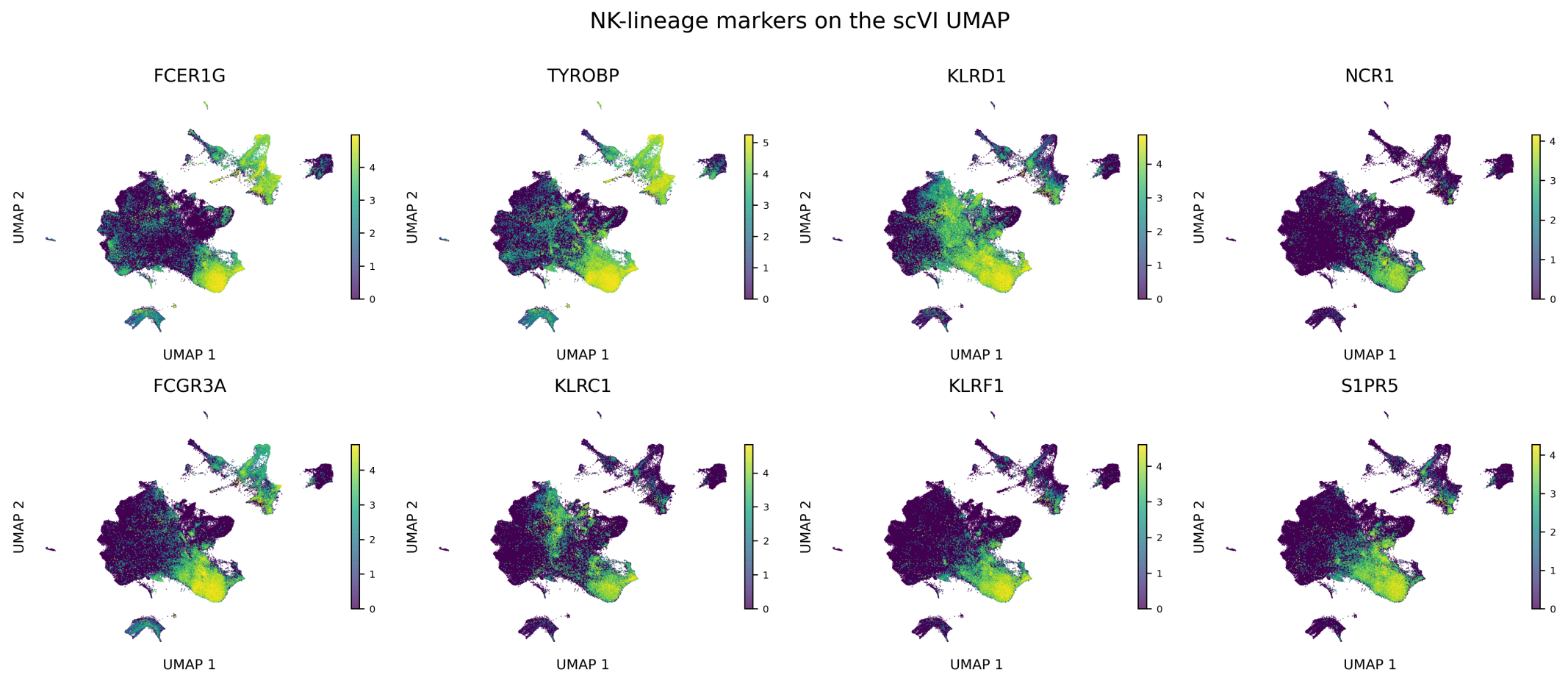
3. T-cell lineage feature plots

T-cell lineage markers on the scVI UMAP



Expression is log1p CP10K within the fixed 4,000 scVI integration features.

4. NK lineage feature plots

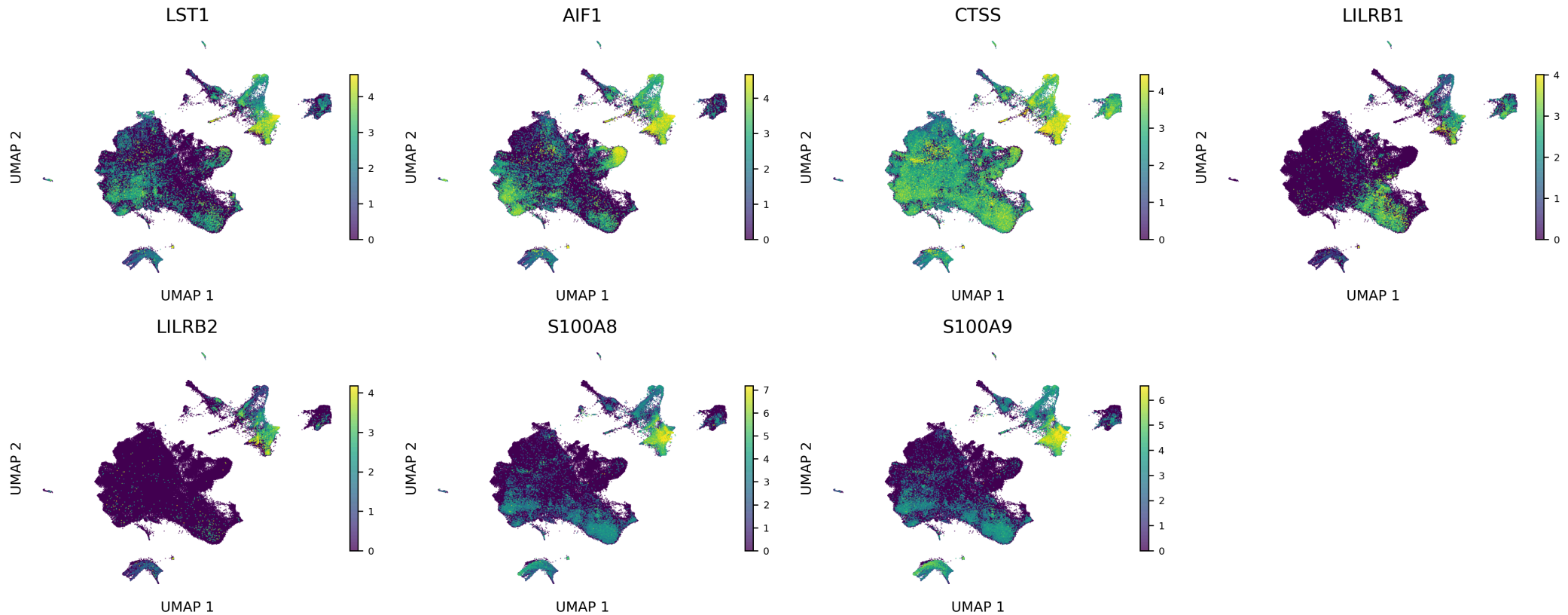


NK evidence requires a multigene program; no single receptor is treated as definitive.

5. Non-T/NK contamination context

FCER1G and TYROBP are not sufficient for NK identity because myeloid cells also express them. The following panel is used only to recognize obvious non-T/NK clusters; these genes are not proposed as gdTAI classifier features.

Myeloid contamination context on the scVI UMAP

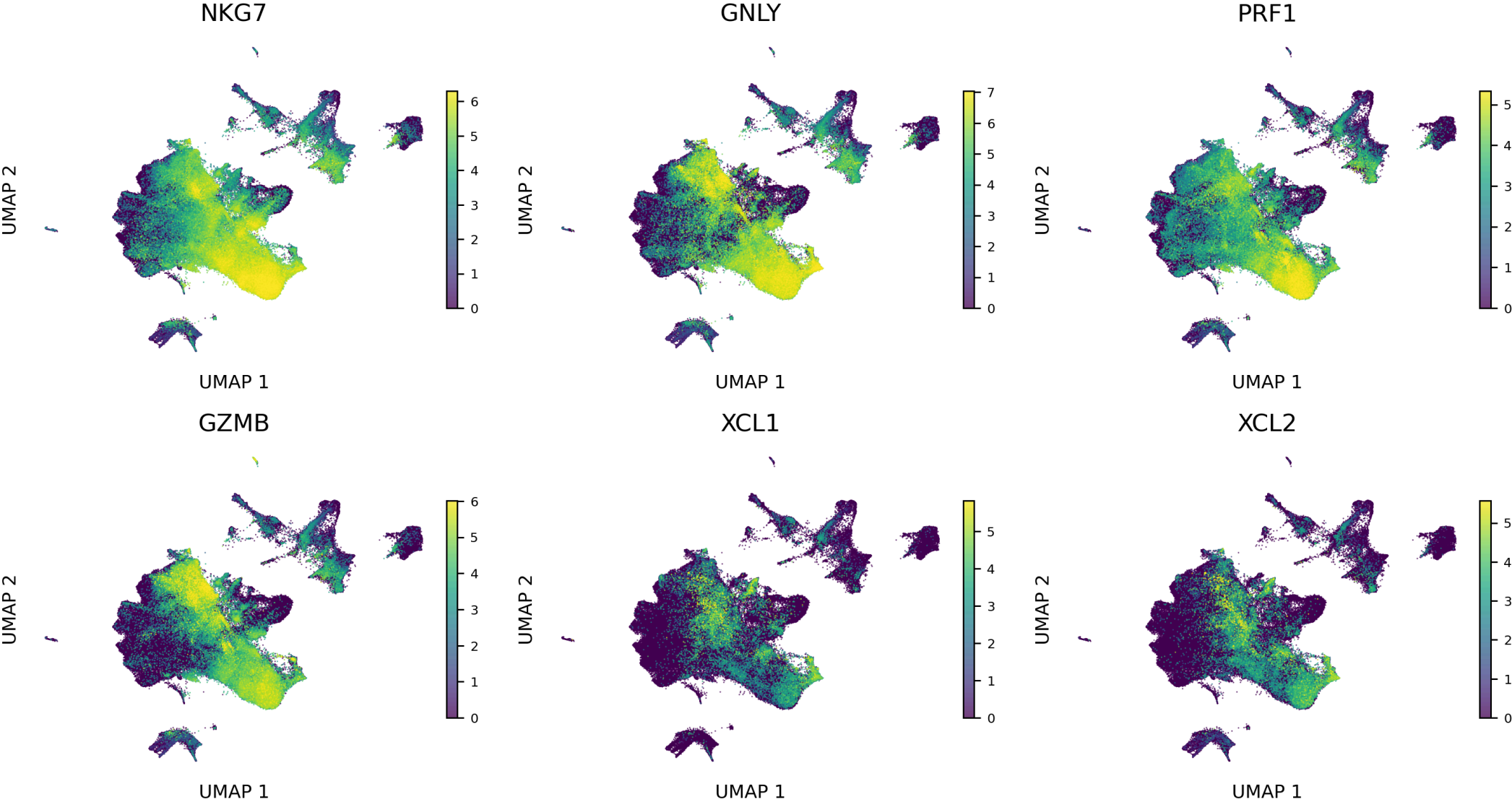


Clusters with broad LST1/AIF1/CTSS/LILRB/S100 support should not be called NK solely from adaptor expression.

6. Shared cytotoxic program

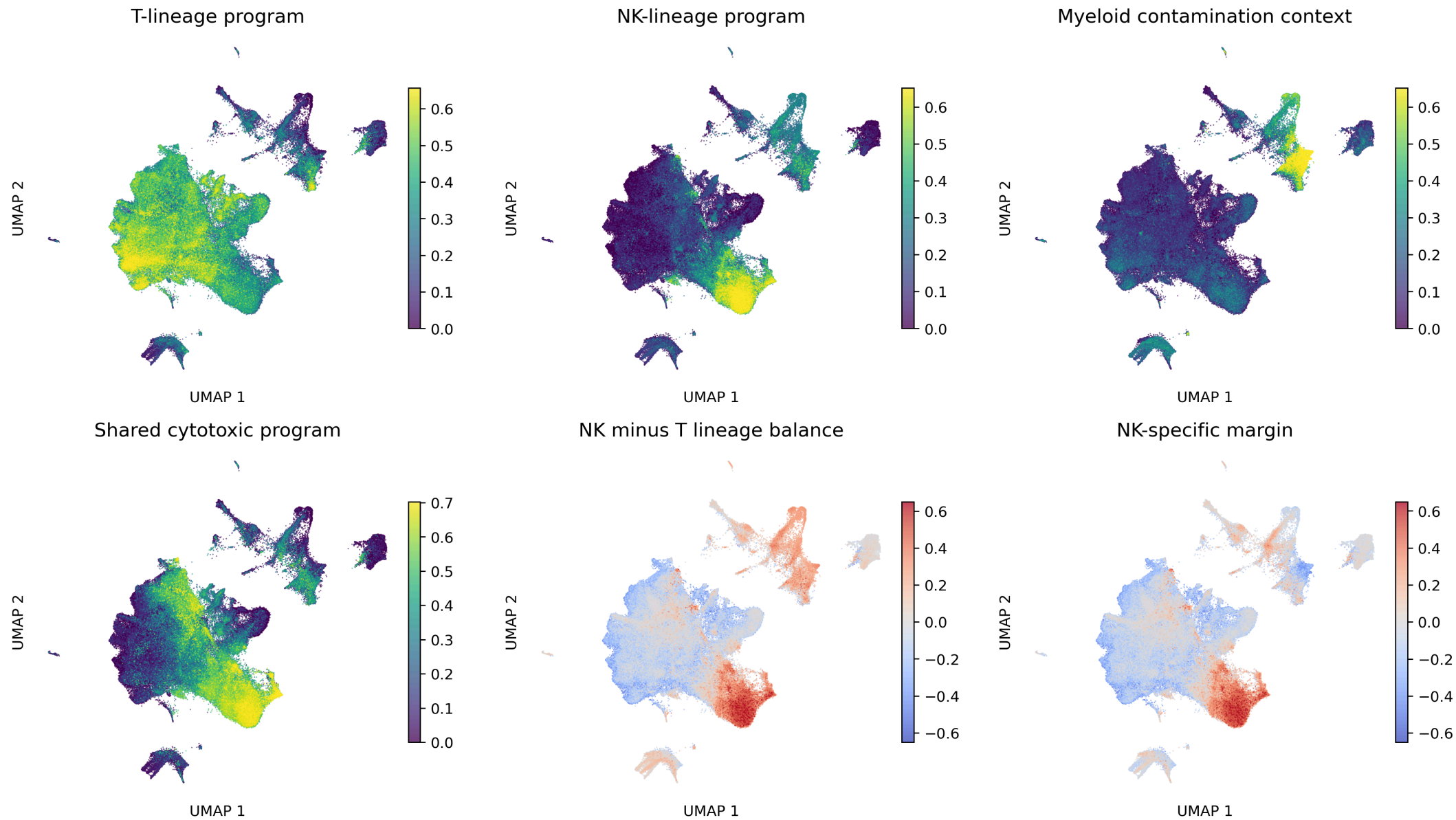
Do not use cytotoxicity as NK truth. NKG7, GNLY, PRF1, GZMB, XCL1, and XCL2 can also be high in CD8 and gamma-delta T cells.

Shared cytotoxic markers on the scVI UMAP



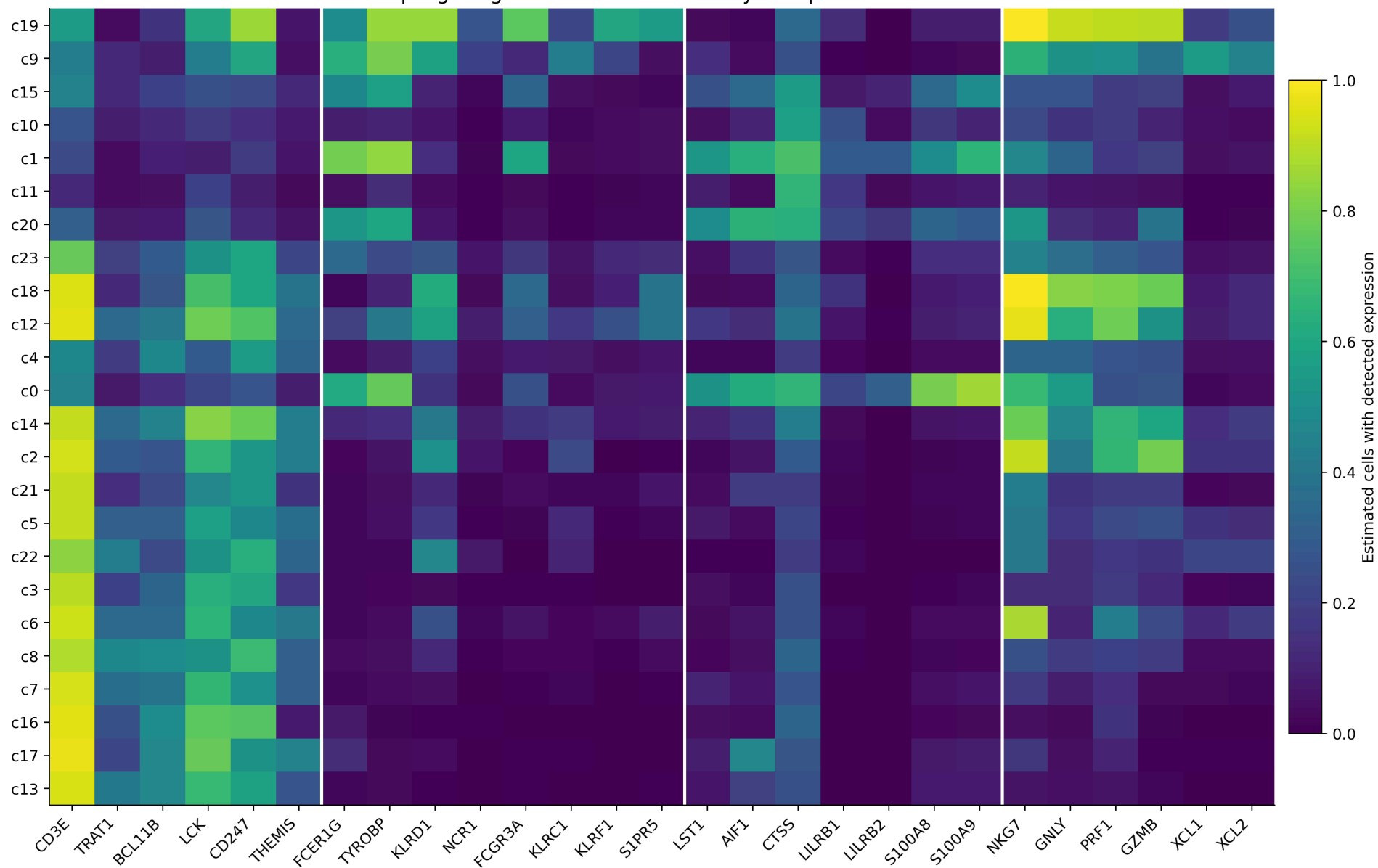
These markers are shown as phenotype context only.

7. Multigene programs and cluster heatmap



Each program is the mean of gene-wise robust-scaled expression. NK-specific margin subtracts the stronger of T-lineage or myeloid-context evidence.

Sampling-weighted marker detection by unsupervised cluster



Sampling-weighted marker detection estimates compensate for the diagnostic sample's source and evidence-role oversampling.

8. Clusters ranked for review

The table ranks clusters by the continuous NK-specific margin: NK lineage minus the stronger of T-lineage or myeloid-context evidence. It does not impose a new hard NK cutoff. Exact anchor, candidate, and source-composition counts are included so expression, annotation provenance, and source bias can be considered together.

cluster	nk_likeness_rank	n_cells	n_primary_nk	n_productive_t	n_eligible_candidates	n_confident_nk	weighted_t_program	weighted_nk_program	weighted_myeloid_program	weighted_cytotoxic_program	nk_minus_t_program	nk_specific_margin	dominant_candidate_source	dominant_candidate_source_fraction
19	1	490300	20237	18316	9162	461	0.254	0.396	0.054	0.522	0.142	0.142	GSE292700	86.7%
9	2	41060	48	686	1139	1	0.157	0.227	0.037	0.309	0.070	0.070	GSE294273	49.3%
15	3	39722	0	2829	14043	0	0.079	0.076	0.119	0.050	-0.003	-0.043	GSE292700	80.2%
10	4	9276	0	907	181	0	0.062	0.026	0.079	0.054	-0.036	-0.053	GSE292700	66.9%
1	5	42063	0	1554	16940	0	0.056	0.207	0.269	0.086	0.151	-0.062	GSE292700	96.9%
11	6	112032	8	5570	5772	0	0.048	0.019	0.093	0.025	-0.029	-0.073	GSE294273	85.8%
20	7	7922	0	515	488	0	0.062	0.095	0.196	0.091	0.033	-0.100	GSE294273	88.3%
23	8	11226	26	4894	39	0	0.292	0.110	0.062	0.168	-0.183	-0.183	GSE292700	97.4%
18	9	236832	262	80173	668	0	0.343	0.134	0.056	0.425	-0.209	-0.209	GSE292700	40.1%
12	10	73248	12	25437	388	0	0.389	0.174	0.067	0.334	-0.215	-0.215	GSE292700	45.1%

9. Recommended decision

Strict core recommendation: retain 461 cells. These cells pass both independent cell-level evidences and fall in cluster 19, which contains 96.1% of all primary NK anchors. Hold the remaining 8 current calls outside cluster 19 rather than using them as strict training NK labels.

No automatic expansion is recommended yet. Cluster 19 contains plausible dropout-rescue tiers, but its eligible candidates are 86.7% from one source, above the 70% source cap. Cluster 9 is a biologically plausible secondary NK-like cluster with better source balance, but it has only 48 primary NK anchors and weak latent-anchor passage. Both require source-resolved review before use.

review_tier	n_cells	n_sources	dominant_source	maximum_source_fraction	recommended_use
A_core_both_cluster19	461	4	GSE292700	96.7%	strict low-weight NK training reference
B_cluster19_latent_only_dropout_review	594	5	GSE292700	91.1%	dropout-rescue review only
C_cluster19_expression_only_distance_review	759	3	GSE292700	91.7%	source/distance review only
D_cluster9_secondary_nk_review	257	3	GSE294273	57.6%	secondary-cluster review only
E_current_confident_outside_core_hold	7	1	GSE292700	100.0%	hold out of strict NK training
F_no_nk_assignment	111209	5	GSE292700	46.7%	do not assign NK

Decision boundary

No H5AD or frozen label manifest is changed by this report. A defensible expansion should require coherent NK-lineage expression, low T-lineage and myeloid support, localization to reproducible unsupervised clusters, and acceptable source composition. Cytotoxicity alone, one receptor, or the historical scANVI label alone is insufficient.

Reproducibility

- Latent SHA-256: 7030b28be885b85a4b24ce2223bb16f7c7288a29195f2512bf7bbba330791ccf
- Partition SHA-256: 90e83ec83986358a015885e02e29d06eacf726d928e02b2e90428d5c09947a63
- Cluster run: global_r0.8_s29
- Marker-expression cache SHA-256: 82ad58d59e336be57a25cabaa48f56e414f674acdb2657103003ff64c2714225
- Script SHA-256: d97058c2464cdc8e45b51d2fa46d9429f50aae25a2e59d41d8218e9cff7a8db6
- H5AD mutation: **none**; classifier fitting: **none**; GitHub push: **none**.